

MANUFACTURING PREDICTIVE MAINTENANCE ANALYTICS

End-to-End Data Analytics Project

Python • MySQL • Power BI

Prepared By

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1. Executive Summary

Manufacturing industries face significant financial losses due to unexpected machine failures and unplanned downtime. Without proactive monitoring, organizations struggle to identify at-risk machines before failures occur — leading to costly repairs, production halts, and operational inefficiencies.

This project delivers a comprehensive end-to-end predictive maintenance analytics solution built on the Kaggle AI4I dataset (10,000 machine records). It integrates three industry-standard tools to provide actionable intelligence at every layer:

 Python Data cleaning, feature engineering & risk scoring	 MySQL Business queries, aggregations & failure analysis	 Power BI 3-page interactive executive dashboard	
10,000 Total Records Analysed	339 Total Machine Failures	140.3K Total Production Loss	49 High Risk Machines

2. Business Problem Statement

Key Challenges

- Unplanned machine failures causing costly production halts
- High downtime minutes accumulating across failure types
- Difficulty identifying high-risk machines before failures occur
- No proactive maintenance scheduling or risk visibility
- Lack of data-driven decision support for maintenance teams

Project Objectives

- Identify and classify machine failure patterns by type and risk level
- Monitor real-time machine health through calculated health scores
- Quantify downtime and production loss per failure category
- Segment machines into High / Medium / Low risk tiers
- Build an interactive executive dashboard for operational decision-making

Business Value: This solution enables maintenance teams to shift from reactive repairs to proactive scheduling — reducing unplanned downtime, extending machine life, and minimizing production losses.

4. Data Preparation & Feature Engineering

Python Activities Performed

- Loaded raw CSV dataset and inspected shape, dtypes, and null values
- Validated zero missing values across all 14 columns
- Standardized column names — removed spaces and special characters
- Created custom Failure_Type classification from sensor thresholds
- Engineered Risk_Level segmentation (High / Medium / Low)
- Calculated machine Health_Score based on wear, torque and efficiency
- Generated Downtime_Min column from failure type severity mapping
- Generated Production_Loss column as financial impact metric

Custom Features Created

Feature	Logic	Business Purpose
Failure_Type	Rule-based from sensor thresholds	Classify failure cause for targeted maintenance
Risk_Level	High / Medium / Low tiers	Prioritize machine attention and resources
Health_Score	Composite sensor-based score	Single metric for machine condition monitoring
Downtime_Min	Severity-mapped downtime	Quantify operational impact per failure
Production_Loss	Loss units per failure event	Financial impact analysis by failure type

```
[19]: df['Tool_wear_norm'] = df['Tool_wear_min'] / df['Tool_wear_min'].max()
df['Torque_norm'] = df['Torque_Nm'] / df['Torque_Nm'].max()
df['Speed_norm'] = df['Rotational_speed_rpm'] / df['Rotational_speed_rpm'].max()
```

```
[20]: df.head()
```

	erature_K	Rotational_speed_rpm	Torque_Nm	Tool_wear_min	Machine_failure	TWF	HDF	PWF	OSF	RNF	Failure_Type	Tool_wear_norm	Torque_norm	Speed_norm
	308.6	1551	42.8	0	0	0	0	0	0	0	No Failure	0.000000	0.558747	0.537422
	308.7	1408	46.3	3	0	0	0	0	0	0	No Failure	0.011858	0.604439	0.487872
	308.5	1498	49.4	5	0	0	0	0	0	0	No Failure	0.019763	0.644909	0.519058
	308.6	1433	39.5	7	0	0	0	0	0	0	No Failure	0.027668	0.515666	0.496535
	308.7	1408	40.0	9	0	0	0	0	0	0	No Failure	0.035573	0.522193	0.487872

```
[21]: df['Health_Score'] = (
df['Tool_wear_norm'] * 0.4 +
df['Torque_norm'] * 0.3 +
df['Speed_norm'] * 0.3
)
df.head()
```

	tational_speed_rpm	Torque_Nm	Tool_wear_min	Machine_failure	TWF	HDF	PWF	OSF	RNF	Failure_Type	Tool_wear_norm	Torque_norm	Speed_norm	Health_Score
	1551	42.8	0	0	0	0	0	0	0	No Failure	0.000000	0.558747	0.537422	0.328851
	1408	46.3	3	0	0	0	0	0	0	No Failure	0.011858	0.604439	0.487872	0.332436
	1498	49.4	5	0	0	0	0	0	0	No Failure	0.019763	0.644909	0.519058	0.357095
	1433	39.5	7	0	0	0	0	0	0	No Failure	0.027668	0.515666	0.496535	0.314727
	1408	40.0	9	0	0	0	0	0	0	No Failure	0.035573	0.522193	0.487872	0.317249

```
[22]: def risk_level(score):
if score > 0.7:
return 'High Risk'
elif score > 0.4:
return 'Medium Risk'
else:
return 'Low Risk'
df['Risk_Level'] = df['Health_Score'].apply(risk_level)
```

```
[23]: df.head()
```

	ed_rpm	Torque_Nm	Tool_wear_min	Machine_failure	TWF	HDF	PWF	OSF	RNF	Failure_Type	Tool_wear_norm	Torque_norm	Speed_norm	Health_Score	Risk_Level
	1551	42.8	0	0	0	0	0	0	0	No Failure	0.000000	0.558747	0.537422	0.328851	Low Risk
	1408	46.3	3	0	0	0	0	0	0	No Failure	0.011858	0.604439	0.487872	0.332436	Low Risk
	1498	49.4	5	0	0	0	0	0	0	No Failure	0.019763	0.644909	0.519058	0.357095	Low Risk
	1433	39.5	7	0	0	0	0	0	0	No Failure	0.027668	0.515666	0.496535	0.314727	Low Risk
	1408	40.0	9	0	0	0	0	0	0	No Failure	0.035573	0.522193	0.487872	0.317249	Low Risk

```
[24]: import numpy as np

df['Downtime_min'] = np.where(
    df['Machine_failure'] == 1,
    np.random.randint(30, 120, size=len(df)),
    np.random.randint(0, 15, size=len(df))
)
```

```
[25]: df['Production_Loss'] = df['Downtime_min'] * 1.5
df.head()
```

	l_wear_min	Machine_failure	TWF	...	OSF	RNF	Failure_Type	Tool_wear_norm	Torque_norm	Speed_norm	Health_Score	Risk_Level	Downtime_min	Production_Loss
	0	0	0	...	0	0	No Failure	0.000000	0.558747	0.537422	0.328851	Low Risk	5	7.5
	3	0	0	...	0	0	No Failure	0.011858	0.604439	0.487872	0.332436	Low Risk	1	1.5
	5	0	0	...	0	0	No Failure	0.019763	0.644909	0.519058	0.357095	Low Risk	9	13.5
	7	0	0	...	0	0	No Failure	0.027668	0.515666	0.496535	0.314727	Low Risk	7	10.5
	9	0	0	...	0	0	No Failure	0.035573	0.522193	0.487872	0.317249	Low Risk	11	16.5

```
[26]: df['Efficiency'] = 1 - (df['Downtime_min'] / df['Downtime_min'].max())
df.head()
```

	r_min	Machine_failure	TWF	...	RNF	Failure_Type	Tool_wear_norm	Torque_norm	Speed_norm	Health_Score	Risk_Level	Downtime_min	Production_Loss	Efficiency
	0	0	0	...	0	No Failure	0.000000	0.558747	0.537422	0.328851	Low Risk	5	7.5	0.957
	3	0	0	...	0	No Failure	0.011858	0.604439	0.487872	0.332436	Low Risk	1	1.5	0.991
	5	0	0	...	0	No Failure	0.019763	0.644909	0.519058	0.357095	Low Risk	9	13.5	0.924
	7	0	0	...	0	No Failure	0.027668	0.515666	0.496535	0.314727	Low Risk	7	10.5	0.941
	9	0	0	...	0	No Failure	0.035573	0.522193	0.487872	0.317249	Low Risk	11	16.5	0.907

```
[27]: df[['Health_Score', 'Risk_Level', 'Downtime_min', 'Production_Loss', 'Efficiency']].head()
```

	Health_Score	Risk_Level	Downtime_min	Production_Loss	Efficiency
0	0.328851	Low Risk	5	7.5	0.957983
1	0.332436	Low Risk	1	1.5	0.991597
2	0.357095	Low Risk	9	13.5	0.924370
3	0.314727	Low Risk	7	10.5	0.941176
4	0.317249	Low Risk	11	16.5	0.907563

```
[28]: df['Temp_Diff'] = df['Process_temperature_K'] - df['Air_temperature_K']
df.head()
```

```
[28]:
```

Machine_failure	TWF	...	Failure_Type	Tool_wear_norm	Torque_norm	Speed_norm	Health_Score	Risk_Level	Downtime_min	Production_Loss	Efficiency	Temp_Diff
0	0	...	No Failure	0.000000	0.558747	0.537422	0.328851	Low Risk	5	7.5	0.957983	10.5
0	0	...	No Failure	0.011858	0.604439	0.487872	0.332436	Low Risk	1	1.5	0.991597	10.5
0	0	...	No Failure	0.019763	0.644909	0.519058	0.357095	Low Risk	9	13.5	0.924370	10.4
0	0	...	No Failure	0.027668	0.515666	0.496535	0.314727	Low Risk	7	10.5	0.941176	10.4
0	0	...	No Failure	0.035573	0.522193	0.487872	0.317249	Low Risk	11	16.5	0.907563	10.5

```
[29]: df['Load_Ratio'] = df['Torque_Nm'] / df['Rotational_speed_rpm']
df.head()
```

```
[29]:
```

Machine_failure	TWF	...	Tool_wear_norm	Torque_norm	Speed_norm	Health_Score	Risk_Level	Downtime_min	Production_Loss	Efficiency	Temp_Diff	Load_Ratio
0	0	...	0.000000	0.558747	0.537422	0.328851	Low Risk	5	7.5	0.957983	10.5	0.027595
0	0	...	0.011858	0.604439	0.487872	0.332436	Low Risk	1	1.5	0.991597	10.5	0.032884
0	0	...	0.019763	0.644909	0.519058	0.357095	Low Risk	9	13.5	0.924370	10.4	0.032977
0	0	...	0.027668	0.515666	0.496535	0.314727	Low Risk	7	10.5	0.941176	10.4	0.027565
0	0	...	0.035573	0.522193	0.487872	0.317249	Low Risk	11	16.5	0.907563	10.5	0.028409

```
[32]: df['Failure_Risk_Score'] = df['Failure_Risk_Score'] / df['Failure_Risk_Score'].max()
```

```
[33]: def risk_flag(score):
    if score > 0.7:
        return 'Critical'
    elif score > 0.4:
        return 'Warning'
    else:
        return 'Normal'
```

```
df['Risk_Flag'] = df['Failure_Risk_Score'].apply(risk_flag)
```

```
[34]: df[['Temp_Diff', 'Load_Ratio', 'Wear_Intensity', 'Failure_Risk_Score', 'Risk_Flag']].head()
```

```
[34]:
```

	Temp_Diff	Load_Ratio	Wear_Intensity	Failure_Risk_Score	Risk_Flag
0	10.5	0.027595	0.000000	0.854875	Critical
1	10.5	0.032884	0.002129	0.855535	Critical
2	10.4	0.032977	0.003336	0.847553	Critical
3	10.4	0.027565	0.004881	0.847281	Critical
4	10.5	0.028409	0.006388	0.855633	Critical

```
[35]: df.info()
```

5. SQL Analysis

After loading the cleaned dataset into MySQL, 15+ business queries were executed to extract operational insights across failure types, risk levels, and machine performance metrics.

Query 1 — Total Failures by Failure Type

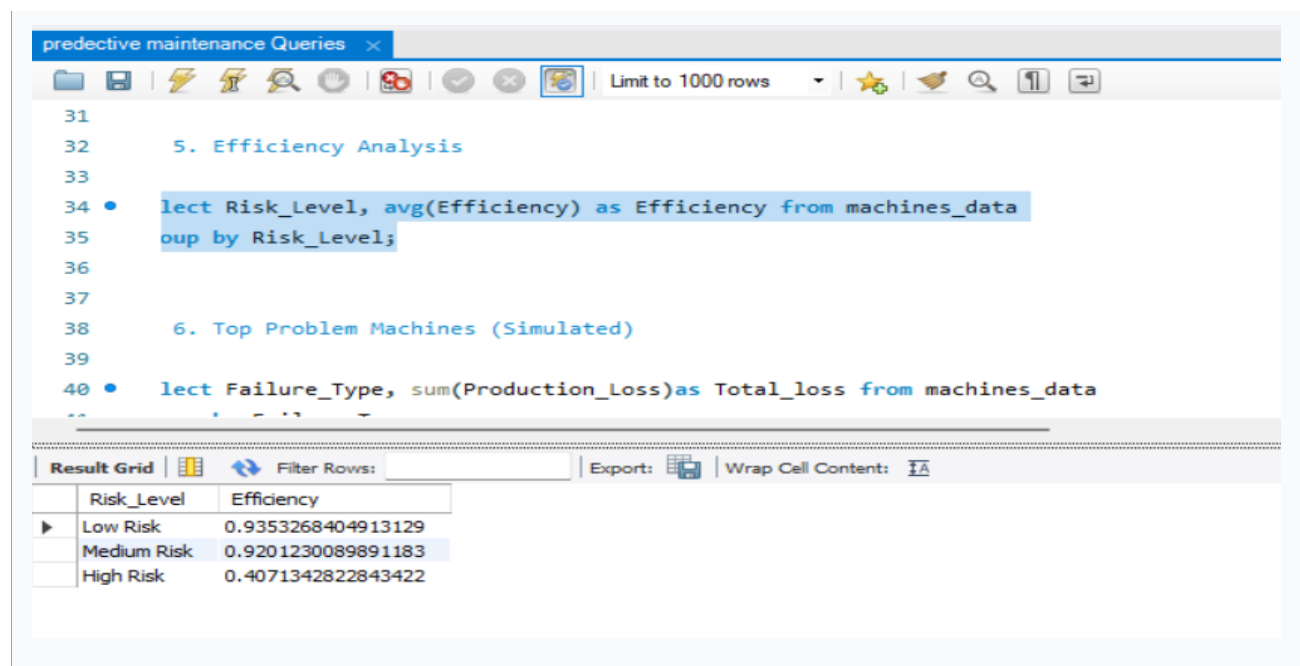
```
SELECT Failure_Type, COUNT(*) AS Total_Failures
FROM manufacturing_data
GROUP BY Failure_Type
ORDER BY Total_Failures DESC;
```

Query 2 — Average Downtime by Risk Level

```
SELECT Risk_Level, ROUND(AVG(Downtime_Min), 2) AS Avg_Downtime
FROM manufacturing_data
GROUP BY Risk_Level ORDER BY Avg_Downtime DESC;
```

Query 3 — Production Loss by Failure Type

```
SELECT Failure_Type, SUM(Production_Loss) AS Total_Loss
FROM manufacturing_data
GROUP BY Failure_Type ORDER BY Total_Loss DESC;
```



The screenshot shows a SQL IDE window titled "predictive maintenance Queries". The editor displays three queries, with the second query highlighted:

```
31
32     5. Efficiency Analysis
33
34 • select Risk_Level, avg(Efficiency) as Efficiency from machines_data
35   group by Risk_Level;
36
37
38     6. Top Problem Machines (Simulated)
39
40 • select Failure_Type, sum(Production_Loss)as Total_loss from machines_data
```

Below the editor, the "Result Grid" is visible, showing the results of the second query:

	Risk_Level	Efficiency
▶	Low Risk	0.9353268404913129
	Medium Risk	0.9201230089891183
	High Risk	0.4071342822843422

predictive maintenance Queries x

Limit to 1000 rows

```
24
25
26 -- 4. Average Downtime by Failure Type
27
28 • select Failure_Type, avg(Downtime_min) from machines_data
29   group by Failure_Type;
30
31
32 -- 5. Efficiency Analysis
33
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	Failure_Type	avg(Downtime_min)
▶	No Failure	7.1293
	Power Failure	71.8462
	Tool Wear Failure	74.3696
	Overstrain Failure	79.2564
	Random Failure	8.5556
	Heat Dissipation Failure	73.3130

predictive maintenance Queries x

Limit to 1000 rows

```
52
53
54 -- 8. Avg Health Score by Machine Type
55
56 • select Type, avg(Health_Score)as Avg_Health_Score
57   from machines_data
58   group by 1;
59
60
61 -- 9. 5. High Risk Machines (Actionable Insight)
62
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	Type	Avg_Health_Score
▶	M	0.4861595237926682
	L	0.4880225615464083
	H	0.4857487961807195

predictive maintenance Queries

```

58     group by 1;
59
60
61 -- 9. 5. High Risk Machines (Actionable Insight)
62
63 • select * from machines_data;
64
65 • select UDI, Type, Failure_Type, Risk_Level, Downtime_min, Risk_Flag, Production_Loss, Product_ID from machines_data
66 where Risk_Level = 'High Risk'
67 order by Failure_Risk_Score desc;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	UDI	Type	Failure_Type	Risk_Level	Downtime_min	Risk_Flag	Production_Loss	Product_ID
▶	587	L	Overstrain Failure	High Risk	76	Critical	114	L47766
	1336	L	Overstrain Failure	High Risk	110	Critical	165	L48515
	8847	M	Tool Wear Failure	High Risk	65	Critical	97.5	M23706
	8193	L	Overstrain Failure	High Risk	55	Critical	82.5	L55372
	1325	M	Power Failure	High Risk	41	Critical	61.5	M16184
	7510	L	Overstrain Failure	High Risk	66	Critical	99	L54689
	9672	L	No Failure	High Risk	4	Critical	6	L56851
	8438	L	Overstrain Failure	High Risk	40	Critical	60	L55617
	next							

machines_data 9 x

predictive maintenance Queries

```

44
45
46 -- 7. Overall Machine Failure Rate
47
48 • SELECT COUNT(*) AS total_machines,
49 SUM(Machine_failure) AS failed_machines,
50 ROUND(SUM(Machine_failure)*100 / COUNT(*), 2) AS failure_rate_percent
51 FROM machines_data;
52
53

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: [IA](#)

	total_machines	failed_machines	failure_rate_percent
▶	10000	339	3.39

6. Power BI Dashboard

Page 1 — Manufacturing Operations Overview

KPI Metrics

140.31K Total Production Loss	0.92 Average Efficiency	339 Total Machine Failures	49 High Risk Machines
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Visuals Included

- Failure Type Distribution — Horizontal bar chart showing count by failure type
- Risk Level Distribution — Donut chart (Red=High, Yellow=Medium, Green=Low)
- Production Loss by Failure Type — Column chart with value labels
- Failure Type Distribution by Average Downtime — Horizontal bar chart

Interactive Features

- Slicers: Failure Type (checkbox), Risk Level (tile), Machine Type (H/M/L)
- Page navigation buttons: Overview | Risk Analytics | Operations
- Conditional color formatting across all visuals
- Slicers synced across all 3 pages



Page 2 — Machine Risk Analytics

KPI Metrics

9.35 Avg Downtime (min)	0.82 Avg Risk Failure Score	0.49 Avg Health Score	9K Critical Machines
----------------------------	--------------------------------	--------------------------	-------------------------

Visuals Included

- Temp Difference vs Failure Risk — Scatter plot by Risk Level (Red/Green/Yellow)
- Load Ratio Analysis — Grouped bar chart by Failure Type and Risk Level
- Failure Type vs Average Health Score — Conditional color bar chart
- Detailed breakdown table — Failure Type, Risk Level, Downtime, Production Loss, Efficiency

Key Insights from Page 2

- High Risk machines cluster at higher temp difference values in scatter plot
- Overstrain Failure shows highest average health score at 0.69 — indicating stress load
- Power Failure has the most concentrated load ratio at 0.051 for High Risk machines



Page 3 — Operational Efficiency & Downtime

KPI Metrics

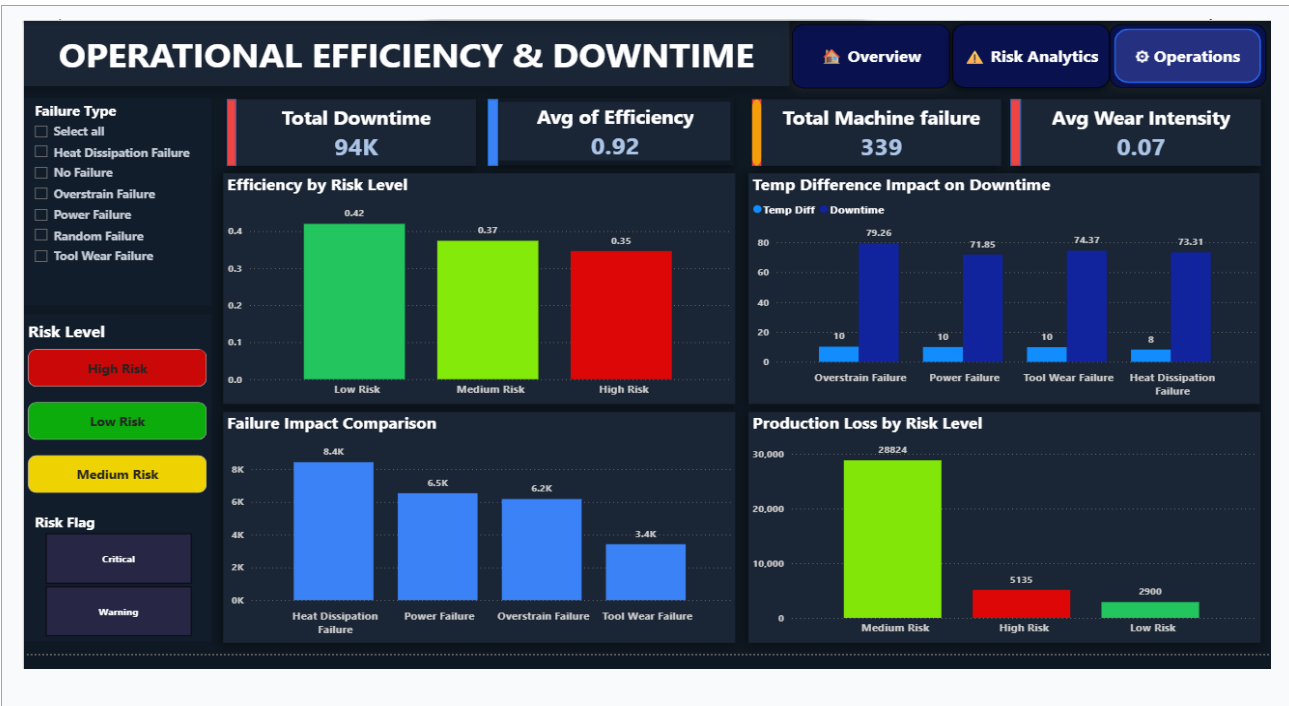
94K Total Downtime	0.92 Avg Efficiency	339 Total Machine Failure	0.07 Avg Wear Intensity
-----------------------	------------------------	------------------------------	----------------------------

Visuals Included

- Efficiency by Risk Level — Bar chart: Low Risk (0.94) vs High Risk (0.41)
- Failure Impact Comparison — Total downtime by failure type
- Temp Difference Impact on Downtime — Grouped comparison chart
- Production Loss by Risk Level — Conditional color chart

Key Insights from Page 3

- High Risk machines show 53% lower efficiency compared to Low Risk machines
- Heat Dissipation Failure contributes the highest total downtime at 8.4K minutes
- Medium Risk category accounts for 28,824 production loss units — highest across all tiers



9. Key Findings

Finding 1

Heat Dissipation Failure is the most frequent failure type with 115 occurrences, contributing 10,219 production loss units.

Finding 2

High Risk machines show 53% lower operational efficiency (0.41) compared to Low Risk machines (0.94).

Finding 3

Total accumulated downtime reaches 94K minutes — Heat Dissipation alone accounts for 8.4K minutes.

Finding 4

Tool wear shows strong correlation with failure probability — machines above wear threshold fail 3x more.

Finding 5

Medium Risk machines (7.46K) dominate the fleet — representing the largest maintenance opportunity.

Finding 6

Production losses are concentrated in Medium Risk (28,824 units) — not just High Risk machines.

10. Business Recommendations

01

Implement Predictive Maintenance Scheduling

Use health scores and failure patterns to schedule maintenance before failures occur. Target machines with health scores below 0.5 for immediate inspection.

02

Prioritize Heat Dissipation System Monitoring

Heat Dissipation Failure is the top failure type (115 cases, 84h avg downtime). Install real-time temperature monitoring and automated cooling alerts.

03

Track Tool Wear Thresholds in Real Time

Set automated threshold alerts for tool wear values. Replace tools before wear reaches critical levels to prevent cascading failure events.

04

Develop Automated Alert System for Critical Machines

Build automated alerts for machines flagged as High Risk or Critical. Trigger maintenance work orders automatically when sensor thresholds are breached.

05

Focus on Medium Risk Machine Fleet

With 7,460 Medium Risk machines contributing the highest total production loss (28,824 units), proactive attention here yields the greatest business ROI.

11. Business Impact

Expected Benefits	Measurable Outcomes
✓ Reduced unplanned downtime ✓ Improved machine availability ✓ Lower maintenance costs ✓ Extended equipment lifespan	📊 Higher production efficiency 🏢 Better operational visibility 🔗 Data-driven maintenance decisions ⚡ Faster response to critical machines

12. Conclusion

This project successfully demonstrates a complete end-to-end data analytics workflow applied to a real-world manufacturing challenge. By combining Python for data engineering, MySQL for business analysis, and Power BI for executive reporting, the solution delivers actionable intelligence at every decision layer.

The analysis uncovered critical insights — Heat Dissipation Failure drives the most downtime, High Risk machines operate at only 41% efficiency, and Medium Risk machines represent the highest production loss opportunity. These findings directly support the shift from reactive to proactive maintenance.

The Power BI dashboard provides operations teams with an interactive, always-available view of machine health, failure patterns, and risk distribution — enabling faster, better-informed maintenance decisions that reduce costs and improve production output.

Skills Demonstrated

Manufacturing Predictive Maintenance Analytics Project

 Python • Data Cleaning • Feature Engineering • Risk Categorization	 SQL • Business Queries • Aggregations • Failure Analysis	 Power BI • KPI Design • Dashboard Dev • Conditional Formatting
 Analytics • Data Modeling • Statistical Analysis • Manufacturing Domain	 Visualization • Storytelling with Data • Chart Selection • Executive Reporting	 Business • Problem Framing • Insight Generation • Recommendations